

RegTrace-Agent: A Context-to-Feedback Framework for Evidence-Grounded Regulatory Response Review

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Abstract

Financial LLM systems are increasingly asked to judge whether an institution’s response actually resolves a prior review request, yet most financial benchmarks still evaluate static QA, extraction, or classification. We introduce RegTrace-Agent, a context-to-feedback-review framework for evidence-grounded response resolution. Given a reviewer request, an organizational response, and revised or corrective artifacts, the framework retrieves context, decomposes obligations, verifies visible gaps, converts adjudicated failures into scalar/category/natural-language feedback, and trains a reusable reviewer policy. We instantiate the framework primarily on SEC comment-letter review, constructing RegTrace-SEC, 472 public EDGAR examples with amended-filing evidence, visible-evidence labels, feedback fields, and later same-topic SEC follow-up signals. We further test cross-regulatory adaptation on CMS-2567 Plans of Correction and an FDA warning-letter stress test, where the framework must reshape the task because public company responses are not observable. Across SEC grouped, temporal, and topic-holdout settings, full natural-language feedback is the strongest reviewer-training signal among handwritten prompting, MIPROv2, and scalar/category GEPA variants; on CMS, domain-specific full-feedback optimization beats generic and scalar/category controls; on FDA, recursive rubric calibration improves a reshaped warning-to-closeout verifier. Qualitative traces show how feedback turns a terse reviewer into an obligation-evidence policy that can reject unsupported revision claims while avoiding over-requirement when a corrective plan or closeout trace is already materially adequate.

1 Introduction

Financial NLP has rapidly moved toward LLM systems for long documents, domain-specific reasoning, and decision support. Models and benchmarks such as BloombergGPT, FinGPT, FinBen,

FinanceBench, FinQA, TAT-QA, and ConvFinQA show both the promise of financial LLMs and the continuing difficulty of grounding answers in financial evidence (Wu et al., 2023; Yang et al., 2023; Xie et al., 2024; Islam et al., 2023; Chen et al., 2021; Zhu et al., 2021; Chen et al., 2022). Recent FinNLP work has similarly emphasized retrieval-augmented financial QA, SEC filing QA, financial intelligence benchmarks, factuality judging, and evasive-answer detection (Lithgow-Serrano et al., 2025; Lai et al., 2025; Klimaszewski et al., 2025; Wu et al., 2025; Al Nuaimi et al., 2025). Most such settings, however, still present a static input and ask for a static answer, label, or report.

Many operational finance and compliance workflows have a different structure. A regulator, auditor, lender, or control function requests a correction, explanation, disclosure, or supporting artifact; the organization responds that it has revised an external document; the reviewer must decide whether visible evidence actually satisfies the request. For example, a company may reply that it revised a filing to address a requested forecast disclosure, while the amended filing only shows combined revenue projections and still omits the requested entity-level assumptions. The task is therefore not to judge whether the response sounds cooperative, but whether the repaired artifact visibly closes the original gap. We call this class of problems evidence-grounded response resolution. It is related to fact verification and document-level legal NLI (Thorne et al., 2018; Koreeda and Manning, 2021), but adds a prior reviewer request, a response that may only claim a revision, revised evidence that must be checked, and textual feedback about what remains missing.

Existing work leaves three limitations for this setting. First, benchmarks rarely ask whether an organization has repaired a documented deficiency. Second, response evaluation often has weak obligation grounding: a response can sound persuasive

086 while amended evidence omits a named disclosure,
087 quantification, exhibit, legal basis, or accounting
088 analysis. Third, prompt and pipeline optimization
089 often discards feedback structure by receiving only
090 scalar success, although review failures are natu-
091 rally described in text: what was requested, what
092 was claimed, what evidence showed, and what gap
093 remained.

094 We study these limitations through U.S. Securi-
095 ties and Exchange Commission (SEC) comment-
096 letter review. SEC staff issue comments request-
097 ing expanded disclosure, clarification, revised ac-
098 counting discussion, removal of unsupported state-
099 ments, or additional exhibits; companies respond
100 in writing, often with statements such as “the dis-
101 closure has been revised accordingly.” After review
102 completion, the SEC releases staff comment letters
103 and company responses through EDGAR. These
104 traces often point to amended filings and some-
105 times include later same-topic SEC follow-up com-
106 ments. Accounting research treats comment let-
107 ters as economically meaningful regulatory signals
108 (Cassell et al., 2013; Bozanic et al., 2017; Baugh
109 and Schmardebeck, 2023; Ryans, 2021); for NLP,
110 they provide observable request-response-evidence-
111 feedback traces.

112 We introduce RegTrace-Agent, a context-to-
113 feedback-review framework for evidence-grounded
114 reviewers. It reconstructs review traces, retrieves
115 revised or corrective context, decomposes requests
116 into obligations, adjudicates visible gaps, com-
117 piles scalar/category/full feedback, and optimizes
118 a guarded reviewer policy. The learned reviewer is
119 not a single hand-tuned prompt; it is the policy in-
120 side a system that gathers evidence, checks obliga-
121 tions, emits a guarded verdict, and uses adjudicated
122 failures to improve the next offline reviewer. This
123 framing is deliberately broader than SEC filings:
124 the same loop applies whenever an organization re-
125 sponds to a reviewer by claiming to fix a document,
126 process, artifact, or remediation plan.

127 This design makes response resolution a
128 strong testbed for language-feedback optimization.
129 Prompt-programming work treats LLM pipelines
130 as declarative programs whose prompts and demon-
131 strations can be optimized (Khattab et al., 2023);
132 GEPA argues that natural-language feedback can
133 expose reusable failure structure that scalar reward
134 discards (Agrawal et al., 2026). In our setting, that
135 structure is regulator-style: a mistake can be ex-
136 plained as a missing requested disclosure, unsup-
137 ported revision claim, absent exhibit, or incomplete

analysis.

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The above limitations motivate three research
questions. RQ1 asks whether public review
traces can be converted into a benchmark for
evidence-grounded response resolution. RQ2
asks whether full natural-language feedback is
a stronger reviewer-training signal than scalar
prompt search and coarse categories. RQ3 asks
whether the resulting decisions generalize across
threads, time, and issue categories, and whether
unresolved decisions are corroborated by later reg-
ulator follow-up.

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We make four contributions. First, we de-
fine evidence-grounded response resolution and
introduce RegTrace-Agent, a context-to-feedback-
review framework for evidence retrieval, obligation
checking, feedback construction, adaptive trace
specification, and reviewer optimization. Sec-
ond, we instantiate it as RegTrace-SEC, a public
EDGAR benchmark with SEC comments, com-
pany responses, amended-filing evidence, labels,
and feedback fields. Third, we show that full
natural-language feedback is the strongest reviewer-
training signal, outperforming handwritten prompt-
ing, MIPROv2, scalar/category GEPA variants,
and a supervised lexical baseline; a guarded ver-
ifier distilled from the learned policy improves
further. Fourth, we provide evidence ablations,
independent-feedback and feedback-component ab-
lations, prompt-evolution analysis, systematic cor-
roboration from real next-round SEC follow-up
comments, and cross-regulatory studies on CMS
and FDA showing that the framework transfers
as an adaptive construction-and-feedback pattern
rather than as a verbatim SEC prompt.

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2 Related Work

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Financial NLP and evidence-grounded evalua-
tion. Financial NLP has moved from broad docu-
ment classification toward long-document reason-
ing, evidence grounding, and compliance-sensitive
decisions. FinQA, TAT-QA, and ConvFinQA intro-
duced financial reasoning questions over reports,
tables, and conversations (Chen et al., 2021; Zhu
et al., 2021; Chen et al., 2022); FinanceBench eval-
uates open-book QA over company documents (Is-
lam et al., 2023). Recent FinNLP work has simi-
larly emphasized RAG over financial documents,
accessible finance-intelligence benchmarks, factu-
ality and causality judging, SEC filing QA, and
evasive-answer detection (Lithgow-Serrano et al.,

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2025; Klimaszewski et al., 2025; Lai et al., 2025; Al Nuaimi et al., 2025; Wu et al., 2025). Our task differs by asking whether retrieved evidence satisfies a prior regulator request.

SEC comment letters and regulatory review. Accounting and disclosure research has studied SEC comment letters as regulatory signals, including remediation costs, disclosure effects, recurring disclosure deficiencies, and textual classification of comment-letter topics (Cassell et al., 2013; Bozanic et al., 2017; Baugh and Schmarbeck, 2023; Ryans, 2021). That literature generally treats comment letters as economic or disclosure events. We instead treat the correspondence as document-grounded supervision for LLM systems. Each review unit contains a staff request, company response, amended filing evidence, and sometimes a next-round SEC follow-up.

Prompt optimization and language feedback. DSPy reframes LLM pipelines as declarative programs whose prompts and demonstrations can be optimized against metrics (Khatab et al., 2023). MIPROv2 and related optimizers primarily use scalar feedback. A broader line studies language feedback, including self-refinement, verbal reinforcement, and critique-based training (Scheurer et al., 2022; Madaan et al., 2023; Shinn et al., 2023). GEPA uses natural-language reflection over execution traces (Agrawal et al., 2026). This distinction is central here because SEC response errors are naturally textual: missing disclosures, unsupported revision claims, absent exhibits, incomplete analysis, or over-demand beyond the SEC request.

3 RegTrace-Agent Framework

3.1 General Review-Trace Setting

RegTrace-Agent applies to workflows where a response claims to satisfy a prior review request by modifying or supplying evidence outside the response itself. Formally, each review trace contains a request q , a response r , a collection of revised or supporting artifacts $\mathcal{D}$, and optional later reviewer feedback u^+ . The system retrieves evidence $e = R(q, r, \mathcal{D})$, extracts obligations $o_{1:m} = O(q)$, and applies a verifier $V(q, r, e, o_{1:m})$ whose output is either resolved or unresolved. The same trace can also produce feedback for learning. A scalar signal says whether the prediction is correct. A category signal names a broad issue or gap type. A full natural-language signal describes the request,

the response action, the visible evidence, and any unmet requirement. This makes the framework applicable beyond SEC review whenever request-response-evidence traces are available.

3.2 Adaptive Trace Construction

The first stage of RegTrace-Agent is a trace-construction agent rather than a fixed SEC preprocessor. For a candidate domain d , it emits a TRACESPEC

$$S_d = (\phi_d, y_d, f_d, a_d, \rho_d),$$

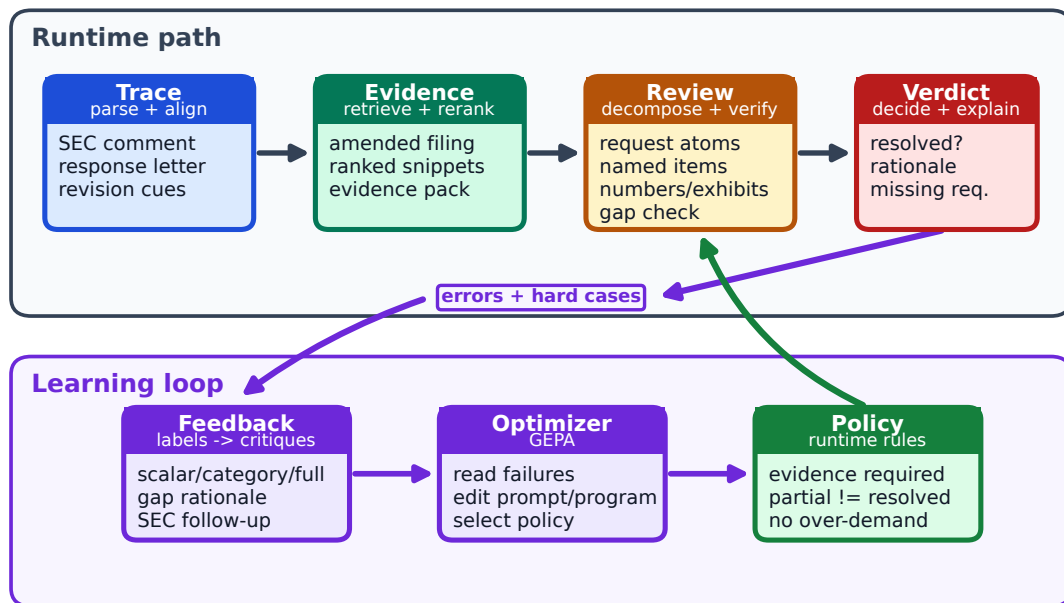
where ϕ_d maps local fields into request, response, evidence, metadata, and external-corroboration roles; y_d defines the review label; f_d defines scalar, category, and full-text feedback channels; a_d lists fields forbidden at test time; and ρ_d is a utility decision. The utility decision scores role observability, evidence availability, label viability, feedback richness, external corroboration, release viability, and optimization suitability. If any core role is weak, the agent must either reshape the task or reject it instead of forcing the SEC schema.

This matters because regulatory corpora expose different trace structures. In SEC review, the appropriate trace is staff comment, company response, amended-filing snippets, visible-evidence label, and later same-topic follow-up. In CMS-2567, the response and corrective artifact are often the same Plan of Correction, so the task becomes plan adequacy rather than amended-document verification. In a probe over FDA warning letters, public warning and closeout letters are available, but public company response letters are largely absent; the construction decision is therefore to reshape the domain into warning-to-closeout resolution tracing or add contrastive controls before treating it as a benchmark. The framework is thus recursive at the data level: construction decisions create feedback fields, feedback improves the reviewer, and reviewer failures can trigger schema or rubric revision.

3.3 SEC Instantiation

Each SEC example contains three test-time fields: `sec_comment`, `company_response`, and `retrieved_snippets` from amended filings. The output is binary. `resolved` means visible amended evidence covers the material request; `unresolved` means a named item, quantification, exhibit, document, legal analysis, accounting analysis, or other

RegTrace-Agent: closed-loop evidence review and feedback optimization



Test-time inputs: SEC comment, company response, retrieved amended evidence. Feedback fields are train/audit only.

Figure 1: RegTrace-Agent architecture. A domain adapter first maps a corpus into request, response, evidence, label, feedback, and optional external-corroboration roles. The runtime path parses a review trace, retrieves or selects evidence, checks obligations, and emits a guarded verdict. The learning loop stores scalar, category, and natural-language feedback, lets GEPA revise the reviewer policy, and reinjects the learned policy into the obligation-review step. At test time the reviewer sees only domain-approved input fields, never adjudication rationales or feedback fields.

285 material requirement remains missing. The label is
 286 a visible-evidence judgment, not a legal determina-
 287 tion or an inference from absence of later follow-
 288 up.

289 3.4 SEC Trace Construction

290 Figure 1 separates the runtime reviewer from the
 291 feedback-learning loop. In the SEC instantiation,
 292 we harvest UPLOAD and CORRESP materials from
 293 EDGAR, group staff comments and company re-
 294 sponses into review threads, and select cases where
 295 the company claims a disclosure was added, re-
 296 vised, removed, or filed elsewhere. We identify
 297 amended filing candidates using filing metadata,
 298 amendment references, page and section cues, and
 299 response text. Candidate filings are drawn from
 300 a window of 10 days before to 60 days after the
 301 response date, with at most 8 candidate filings per
 302 row. A local lexical ranker retrieves top chunks,
 303 and an LLM reranker/adjudicator evaluates the top
 304 3 snippets. From 589 revision-claim cases, 588
 305 have candidate filings and snippets; 472 remain
 306 after requiring visible evidence that directly or par-
 307 tially addresses the SEC request. We then adjudi-

cate whether visible evidence satisfies the request
 and construct labels plus feedback fields.

The retrieval component gives the agent evi-
 dence to inspect. The adjudication and feedback
 layer gives it something to learn from: what the
 reviewer asked for, what the company claimed to
 do, what the visible evidence supports, and what
 remains missing. These adjudication fields are not
 shown as ordinary test-time inputs. They are used
 for optimization feedback, audits, and analysis.

318 3.5 Scale, Categories, and Splits

RegTrace-SEC contains 472 examples from 109
 SEC review threads and 95 companies. The bench-
 mark focuses on difficult revision-claim cases, and
 the label distribution is operationally realistic: 153
 examples are resolved, and 319 are unresolved.
 Table 1 summarizes the corpus and split regimes.

The corpus covers seven issue categories and
 eight visible-evidence gap types; Appendix E gives
 the full distribution. We use grouped, temporal, and
 topic-holdout regimes. The grouped split holds out
 entire review threads (229/104/139 train/dev/test).
 The temporal split trains on 2022–2023 and tests

Quantity	Count
Examples	472
SEC review threads	109
Companies	95
Resolved / unresolved	153 / 319
Examples with verified follow-up	236
Issue categories	7
Visible-evidence gap types	8

Table 1: Benchmark scale. Issue categories include other disclosure issues, risk factors, SPAC disclosure, revenue recognition, crypto, non-GAAP, and cybersecurity.

on 2024–2025 (264/64/144). The topic-holdout split tests transfer to held-out issue categories (279/68/125), including crypto and non-GAAP cases absent from training categories.

3.6 From Labels to Feedback

A standard benchmark would stop at binary labels. Our construction preserves the textual structure of each decision. For every example, adjudication records the material SEC request, the company action, evidence relevance, supporting evidence or resolution basis, visible unmet requirement, and a short rationale. This lets us compare increasingly rich optimization signals:

- scalar: correct or incorrect;
- category: correct/incorrect plus coarse issue or gap type;
- full: natural-language feedback describing the SEC request, company action, evidence coverage, and unmet requirement.

Full feedback can state that the evidence discusses combined revenue projections but fails to disclose entity-level operating costs, product pricing, gross margins, and forecast limitations. This is the reusable decision rule that scalar metrics discard.

4 Methods

4.1 Prediction Setup

The prediction model receives the SEC comment, company response, and raw retrieved amended-filing snippets. It returns a binary label and a short rationale. Explicit resolved is scored as resolved. unresolved, partially_resolved, and other insufficient labels are normalized to unresolved. The rationale is required for inspection but not directly rewarded.

4.2 Optimization Ladder

We compare reviewer-training signals rather than evidence access. Baseline prompt is a handwritten reviewer instruction. MIPROv2 is scalar prompt optimization. GEPA-scalar uses reflective prompt optimization with correctness-only feedback. GEPA-category adds coarse category feedback. GEPA-full uses full natural-language adjudication feedback. We also include TF-IDF, a frozen sentence-encoder scoring head, a Jev typed decision gate, and an independent-feedback GEPA ablation for reviewer-facing robustness.

All methods receive the same evidence at test time, isolating the optimization signal rather than evidence access. The key comparison is whether natural-language feedback adds value beyond prompt search itself.

4.3 Implementation Details

All main prompt-optimization runs use openai/gpt-4o-mini as the frozen prediction model with temperature 0 and 500 output tokens. GEPA uses the same model as the reflection model with temperature 0.7 and 1500 reflection tokens, seed 17, max_metric_calls=150, reflection minibatch size 4, Pareto candidate selection, and 4 evaluation threads. MIPROv2 uses the same prediction and prompt models, 8 trials, 4 prompt candidates, and at most 4 labeled and 4 bootstrapped demonstrations. The frozen encoder scorer uses all-MiniLM-L6-v2 with a logistic head over request, response, evidence, and request-evidence matching features. The Jev gate uses a choice question over resolved/unresolved. The evidence reranker/adjudicator also uses gpt-4o-mini at temperature 0. Model aliases rather than dated provider snapshots are recorded in the released artifacts.

4.4 Optimization as Feedback-Conditioned Search

We view each method as search over a prompt or prompt program p . Given an input x and label y , the frozen LLM defines $\hat{y}_p(x) = f_\theta(x; p)$, and optimization seeks a prompt with high validation reward. The methods differ in what they observe after an error. Scalar optimization observes only correctness. Category feedback adds a coarse gap type. Full feedback adds a short critique describing the request, company action, visible evidence, and unmet requirement.

Input	Acc.	Ma-F1	Res-F1	Unres-F1
Response only	0.712	0.575	0.333	0.817
+ raw snippets	0.784	0.748	0.651	0.844
+ oracle summary	0.964	0.956	0.938	0.975

Table 2: Evidence ablation. Adding raw amended-filing snippets raises macro-F1 by 17.3 points. The oracle summary is an upper bound, not a deployable baseline.

The intuition is information-theoretic. Let C be the latent cause of failure, such as missing quantification or unsupported revision claim. Full feedback should help when it lowers uncertainty about C , because the optimizer can propose a more targeted semantic edit. This is not a general guarantee: a noisy or misaligned language channel can hurt. The hypothesis of RegTrace-Agent is that review traces are unusually well suited to language feedback because the failure causes are naturally written as reviewer critiques.

4.5 What the Optimizer Sees

During optimization, each candidate prompt is evaluated on development examples. When a prediction is incorrect, the metric returns feedback in one of three forms:

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scalar: incorrect
category: incorrect;
gap_type=missing_specific_disclosure
full: incorrect. The SEC requested
X. The company claimed Y. The visible
evidence shows Z, but does not show
required item W.
```

Full feedback summarizes the adjudicated relationship between the SEC request, response, and retrieved evidence for optimization examples only. It does not reveal hidden evidence at test time; any advantage must come from a better learned checklist for visible-evidence review.

5 Experiments

5.1 Evidence Ablation

We first test whether amended-filing evidence matters. Table 2 evaluates a response-only condition, a raw-evidence condition, and an oracle evidence-summary condition on the grouped held-out test set.

Response-only prompting cannot verify whether a revision claim is supported. Raw snippets nearly double resolved F1, while the oracle summary shows that much of the remaining difficulty is interpreting snippets against a specific request.

5.2 Main Context-to-Feedback Results

Table 3 reports the main reviewer-training ladder on the frozen grouped split. All methods receive the same test-time context.

The majority baseline has high accuracy because unresolved examples dominate, but zero resolved F1. The handwritten prompt is conservative: it has perfect unresolved precision but misses most gaps. MIPROv2 improves substantially, GEPA-scalar improves recall, and GEPA-category is too coarse in this run. GEPA-full is strongest overall, improving macro-F1 by 10.4 points over MIPROv2 and 8.1 points over GEPA-scalar. This shows that full feedback is not merely adding more text; it trains the reviewer to compare requested obligations against visible evidence.

Table 4 addresses two alternate explanations and one deployment question. A TF-IDF logistic regression classifier trained on the same raw evidence reaches only 0.523 macro-F1, so lexical supervision does not explain the GEPA-full gain. A frozen sentence encoder with a trained logistic scoring head reaches 0.667 macro-F1 without generation, and a Jev typed decision gate reaches 0.722, showing that RegTrace-Agent can support cheap triage scorers, but still trails the full feedback-trained reviewer. On the same grouped held-out test set ($N = 139$), the guarded reviewer improves over a monolithic prompt under both gpt-4o-mini and gpt-5.4-mini, suggesting that obligation-evidence structure matters across backbones. To test possible feedback leakage, we re-generate full textual feedback with an independent gpt-5.4-mini writer that sees only test-time fields plus the gold label, not the original adjudication rationales; GEPA with this feedback reaches 0.750 macro-F1, nearly matching the original 0.756.

We also compute five-fold out-of-fold results over review-thread groups. Mean macro-F1 is 0.698 for GEPA-full, 0.647 for MIPROv2, 0.609 for GEPA-scalar, 0.545 for GEPA-category, and 0.446 for the handwritten baseline. The fold standard deviations are nontrivial (0.151 for GEPA-full and 0.121 for MIPROv2), so we treat the single grouped split as a headline comparison and the out-of-fold analysis as stability evidence rather than as a deterministic margin claim.

5.3 Why Full Feedback Helps

The advantage of GEPA-full is not simply risk aversion. Jev reaches very high unresolved recall

Method	Acc.	Macro-F1	Res-F1	Unres-F1	Unres-P	Unres-R
Majority unresolved	0.691	0.409	0.000	0.817	0.691	1.000
Baseline prompt	0.424	0.402	0.518	0.286	1.000	0.167
MIPROv2	0.655	0.652	0.619	0.684	0.929	0.542
GEPA-scalar	0.705	0.675	0.577	0.773	0.824	0.729
GEPA-category	0.597	0.595	0.562	0.627	0.870	0.490
GEPA-full	0.784	0.756	0.674	0.839	0.867	0.812

Table 3: Frozen grouped-split leaderboard. GEPA-full gives the best macro-F1 and unresolved F1 under the same raw-evidence test-time context.

Check	Condition	Macro-F1
Lexical baseline	TF-IDF, response only	0.523
Lexical baseline	TF-IDF, raw snippets	0.523
Scoring head	Frozen encoder, raw snippets	0.667
Decision gate	Jev typed choice, raw snippets	0.722
Reviewer structure	4o-mini monolithic / guarded	0.638 / 0.808
Reviewer structure	5.4-mini monolithic / guarded	0.612 / 0.732
Feedback source	GEPA-full, original feedback	0.756
Feedback source	GEPA-full, independent feedback	0.750
Diagnostic upper bound	TF-IDF, oracle evidence summary	0.858

Table 4: Robustness checks on the grouped held-out test set unless noted. The raw-snippet TF-IDF baseline shows that lexical supervision does not explain the main result. Frozen encoder and Jev scorers are cheap triage components. Multi-LLM reviewer runs show that guarded obligation-evidence structure improves over monolithic prompting under two backbones. The independent-feedback ablation shows that GEPA-full does not depend on reusing the original adjudication rationales.

Reviewer	Macro-F1	Res-F1	Unres-F1
Monolithic reviewer	0.645	0.614	0.676
Obligation-guided reviewer	0.619	0.613	0.624
GEPA-full optimized prompt	0.756	0.674	0.839
RegTrace guarded verifier	0.801	0.727	0.874

Table 5: Reviewer-structure ablation. The guarded verifier is a policy-distillation prototype derived from the obligation-evidence policy exposed by GEPA-full prompt evolution.

and offline optimization, so it is not the cheapest runtime gate. It can also become over-skeptical when evidence substantially satisfies a request but does not present the answer in the exact form the prompt expects. We therefore view Jev and frozen encoders as high-throughput first-pass gates, and GEPA-full as the deeper reviewer for difficult or high-impact cases.

5.4 Reviewer-Structure Ablation

The optimization ladder tests feedback richness. We also ask whether reviewer structure matters. Table 5 compares a monolithic judgment, an obligation-guided judgment, and a guarded verifier that requires every material obligation to be visibly satisfied, with GEPA-full as the optimized-prompt reference.

Simply listing obligations is not enough; the gain comes from the guard that requires visible support for each material request. We report this as a policy-distillation prototype rather than the primary optimizer comparison, since it adds human-designed structure after prompt-evolution analysis.

5.5 Feedback Component Ablation

Table 6 holds test-time input fixed and changes only optimization-time feedback.

Request-action-only feedback induces aggressive skepticism and the highest unresolved recall, but sacrifices resolved F1. Full feedback teaches both sides of the boundary: unmet requirements for unresolved cases and supporting evidence for

(0.958) but lower resolved F1 (0.576), making it useful as a triage gate that flags likely unresolved responses. GEPA-full has lower unresolved recall but higher macro-F1, because it better balances skepticism with evidence sufficiency. It learns when a company revision claim is unsupported, but also when visible amended evidence already satisfies the SEC request.

The mechanism is visible in the feedback channel. Scalar feedback says only that a decision was wrong; category feedback names a coarse gap. Full feedback states what the SEC requested, what the company claimed, what the amended evidence showed, and what requirement remained unmet. This gives the optimizer a reusable review trace rather than a bare reward. The resulting prompt evolution shifts from generic caution to obligation-level comparison: identify material SEC request elements, inspect amended evidence for named disclosures, quantification, exhibits, and legal or accounting analysis, and distinguish partial disclosure from resolution.

The trade-off is cost and calibration. GEPA-full requires adjudicated natural-language feedback

Feedback	Acc.	Macro-F1	Res-F1	Unres-F1	Unres-P	Unres-R
Full feedback	0.784	0.756	0.674	0.839	0.867	0.812
No evidence rationale	0.712	0.682	0.583	0.780	0.826	0.740
No unmet requirement	0.741	0.708	0.609	0.806	0.833	0.781
Request + action only	0.784	0.724	0.595	0.853	0.806	0.906

Table 6: Feedback component ablation. Request-action feedback induces aggressive unresolved detection, while full feedback gives the best macro-F1 by calibrating skepticism with evidence rationales.

resolved cases.

5.6 Temporal and Topic Generalization

Table 7 shows that the optimized prompt transfers beyond the grouped split. In the temporal setting, GEPA-full trains/tunes on 2022–2023 and tests on 2024–2025 material. In the topic setting, test examples include crypto and non-GAAP cases absent from training issue categories.

The topic-holdout result argues against a curated-topic explanation. Per-topic breakdowns show GEPA-full improves over MIPROv2 on crypto, non-GAAP, revenue recognition, risk factors, and broad other-disclosure cases, with largest gains in revenue recognition and non-GAAP.

6 Cross-Regulatory Adaptation: CMS-2567

We next ask whether RegTrace-Agent is only a curated SEC benchmark or whether the same context-to-feedback-review pattern can be instantiated in a different regulatory domain. We use CMS-2567 nursing-home Statements of Deficiencies and Plans of Correction. The object of review is different from SEC filings: a facility receives a deficiency narrative, submits a Plan of Correction (POC), and the reviewer asks whether the plan materially addresses resident-specific correction, affected population, systemic prevention, monitoring, timing, and accountability. The test-time input is deficiency_text, plan_of_correction_text, ftag, and scope_severity. Feedback-only fields record covered elements, missing or weak elements, and a reviewer rationale; correction-outcome metadata is hidden and not used as a gold label because most cases eventually become corrected.

This experiment is intentionally framed as adaptation, not direct zero-shot transfer. The SEC optimized prompt checks amended disclosure against named SEC requests. CMS POC review requires a different calibration: a plan can be adequate without proving that every future monitoring step has already succeeded, but it is not adequate if it merely

lists education and audits while omitting a cited resident-specific or systemic deficiency.

We construct a 500-example CMS pilot benchmark with grouped train/dev/test splits by source document (270/90/140 examples). Labels map adequate to adequate and both partial and inadequate to not_adequate. The grouped test set contains 37 adequate and 103 not-adequate examples. These labels are LLM-adjudicated and therefore weaker than the SEC benchmark; we use the study to test framework portability rather than to claim a fully validated second benchmark.

Table 8 shows the central pattern. A direct generic CMS prompt is already strong (0.694 macro-F1), which means POC adequacy is closer to ordinary regulatory reading than SEC amended-evidence verification. The SEC-style structured reviewer, however, is too conservative: it predicts only two adequate cases and reaches 0.449 macro-F1 despite high nominal accuracy from class imbalance. This is a useful negative result. RegTrace-Agent does not mean transferring a frozen SEC checklist. It means re-instantiating the same loop with domain-appropriate feedback.

Under that adaptation, GEPA-full improves the minimal CMS seed from 0.642 to 0.723 macro-F1, beats scalar feedback by 5.9 points, and beats the direct generic prompt by 2.9 points. The gain is not pure skepticism. GEPA-full predicts fewer adequate cases than the generic prompt, but it also preserves high adequate recall while improving not-adequate F1. The full feedback teaches the reviewer when standard POC language is enough and when it is boilerplate that does not address the cited deficiency.

What transfers. The reusable part is not a prompt string. It is the context-to-feedback mechanism: identify the reviewer request or deficiency, map the response to material obligations, use adjudicated text to expose covered and missing elements, and optimize a reviewer around that feedback. SEC review learns “do not accept a revision

Split	Method	Acc.	Macro-F1	Unres-F1	Unres-R
Time	Baseline	0.451	0.434	0.336	0.206
	MIPROv2	0.521	0.519	0.489	0.340
	GEPA-scalar	0.653	0.650	0.679	0.546
	GEPA-full	0.826	0.791	0.877	0.918
Topic	Baseline	0.440	0.440	0.426	0.274
	MIPROv2	0.576	0.565	0.635	0.484
	GEPA-scalar	0.528	0.524	0.569	0.411
	GEPA-full	0.784	0.734	0.849	0.800

Table 7: Temporal and topic-holdout generalization. GEPA-full transfers substantially better than scalar optimization.

Method	Seed	Feedback	Acc.	Macro-F1	Adeq-F1	Not-Adeq-F1
Direct generic prompt	generic	none	0.707	0.694	0.631	0.757
Minimal seed	minimal	none	0.650	0.642	0.588	0.696
GEPA-scalar	minimal	correctness only	0.671	0.663	0.610	0.716
GEPA-category	minimal	coarse category	0.643	0.635	0.583	0.688
GEPA-full	minimal	written POC feedback	0.743	0.723	0.647	0.798
Structured SEC seed	SEC-style	none	0.736	0.449	0.051	0.847
Structured GEPA-full	SEC-style	written POC feedback	0.771	0.584	0.304	0.863

Table 8: CMS-2567 adaptation on the grouped 140-example test set. The SEC-style checklist does not transfer verbatim: it predicts almost everything as not adequate. Starting from a lightweight CMS reviewer, full written feedback beats scalar/category feedback and a direct generic prompt.

claim unless amended evidence contains the named disclosure.” CMS review learns “substantial adequacy is enough when the plan covers immediate correction, affected population, systemic prevention, monitoring, and timing, but generic education/audit language is not enough when a cited deficiency component is omitted.”

7 Third-Domain Stress Test: FDA Warning Letters

The CMS study tests positive adaptation. We also test whether the construction layer can avoid a bad adaptation. FDA warning-letter data provides public warning letters and closeout letters, but public company response letters are almost absent in our snapshot. A fixed SEC-style framework would force this into a request-response-evidence task and produce a weak benchmark. RegTrace-Agent instead emits a TRACESPEC decision of reshape-before-benchmark: the observable task is warning-to-closeout trace verification, not response adequacy.

We construct a minimal FDA contrastive benchmark from 400 linked warning-closeout pairs. Each warning receives one positive candidate, its true linked closeout letter, and one hard negative closeout letter, preferably sampled from the same product and subject. This yields 800 examples with grouped train/dev/test splits by warning URL

Reviewer	Acc.	Ma-F1	Mat. F1	Wrong F1
Generic	0.919	0.918	0.912	0.925
RegTrace strict	0.881	0.880	0.865	0.894
RegTrace calibrated	0.950	0.950	0.947	0.952

Table 9: FDA warning-to-closeout trace verification on a 160-example held-out test set using gpt-4o-mini. The strict RegTrace reviewer catches all wrong closeouts but over-rejects true closeouts. A domain-calibrated reviewer preserves mismatch detection while recovering true matches.

(480/160/160). The held-out test set is balanced, and 357 of 400 negatives in the full benchmark are same-product and same-subject wrong closeouts.

Table 9 shows the recursive behavior we want from the framework. The first strict reviewer is too skeptical: it catches every mismatched closeout but rejects 19 true matches, often because FDA closeout letters are short and do not repeat every violation. The calibrated reviewer changes the rubric, not the hidden data. It treats firm identity, CMS/reference number, warning date, product or violation family, issuing office, and explicit corrective-action evaluation as the trace criteria, while not requiring the closeout to restate the full warning letter. This improves macro-F1 from 0.880 to 0.950 and also exceeds the generic prompt. FDA therefore serves as a compact third-domain demonstration of adaptive construction: when a corpus

Group	N	D/P	Rate
Unres. + f/u	160	109	68.1%
Res. + f/u	76	7	9.2%

Table 10: Real follow-up corroboration. Later same-topic SEC comments directly or partially corroborate unresolved visible-evidence labels far more often than resolved labels.

lacks a response channel, RegTrace-Agent reshapes the task and then calibrates the reviewer to the new document genre.

8 External SEC Follow-Up Corroboration

Visible-evidence labels come from an adjudication pipeline, so we test whether they align with later regulator behavior. Verified same-topic next-round SEC follow-up is not the primary label, but it is a useful corroboration signal because it can restate or narrow an unresolved gap.

We additionally audit all 472 examples with an independent gpt-5.4-mini auditor that receives only the SEC comment, company response, and raw retrieved evidence snippets, not the benchmark label, feedback fields, evidence summaries, missing-requirement notes, or later SEC follow-up. The audit agrees with the benchmark label in 345 of 472 cases, reaching 0.731 agreement accuracy and 0.715 macro-F1. We report this as an audit of visible-evidence label consistency, not as expert legal/accounting validation.

Among 236 examples with verified follow-up, unresolved cases are corroborated at 68.1%, while resolved cases are corroborated at only 9.2% (Table 10). Follow-up availability itself is balanced across labels (49.7% for resolved and 50.2% for unresolved; Fisher exact $p = 1.0$), so this is not merely an availability artifact.

Following reviewer-style paired analysis, we also treat later follow-up as a noisy external reference on a clean 220-example subset. The benchmark label reaches 0.756 macro-F1 against this signal, compared with 0.590 for five-fold out-of-fold GEPA-full predictions and 0.460 for the handwritten baseline. We therefore use follow-up primarily as label-validity evidence, not as proof that every GEPA-full win is separately corroborated beyond the unresolved base rate.

In one expense-disclosure case, the SEC asked for material components and trend discussion. The company claimed a revision, but retrieved evidence showed only a line item and totals; GEPA-full

predicted unresolved, and the next-round SEC follow-up again requested the component breakout. SPAC transaction-background cases show the same pattern for missing negotiation details.

9 Prompt Evolution and Case Studies

9.1 From Terse Prompt to Reviewer Policy

Prompt evolution is the most visible evidence that RegTrace-Agent is not simply tuning a label instruction. In both domains, the seed prompt is deliberately weak. SEC starts from: “Classify whether a company resolved an SEC comment using the visible evidence.” CMS starts from: “Classify whether a CMS Plan of Correction adequately resolves the deficiency.” Full feedback then rewrites these seeds into different reviewer policies because the feedback describes different kinds of gaps.

The component analysis in Table 12 shows the same pattern quantitatively. Scalar and category feedback learn generic caution: company claims need support. Full feedback more consistently induces named-request matching and missing-item reasoning, shifting the policy from generic skepticism to obligation-level comparison: request $\rightarrow$ response action $\rightarrow$ visible evidence or correction plan $\rightarrow$ remaining gap.

9.2 What the Feedback Looks Like

The feedback channel is not a hidden test-time input. It is an optimization-time critique. A SEC unresolved example can return: the SEC requested individual-company forecasts and material assumptions; the company claimed to revise the forecast section; the amended evidence shows combined revenue projections; operating costs, pricing, margins, and forecast limitations remain missing. A CMS not-adequate example can return: the plan covers refrigerator temperature logs but does not address the same deficiency narrative’s fall-risk care-plan failure or RN coverage/PBJ issue. These critiques are exactly the information a scalar reward loses.

9.3 SEC Case: Combined Forecasts Are Not Entity-Level Assumptions

One SEC comment asked whether the board received VSee and iDoc projections on an individual-company basis and requested entity-level assumptions, including growth rates, operating costs, product pricing, gross margins, and forecast limitations. The company responded that it revised the

Policy	What the prompt learns	Representative optimized language
SEC seed	Generic evidence check.	“Classify whether a company resolved an SEC comment using the visible evidence.”
SEC GEPA-full	Compare named SEC requests against amended evidence; distrust unsupported revision claims; find missing disclosures, quantification, exhibits, and legal/accounting analysis.	“Carefully compare the SEC request against the company’s response and the amended evidence” and look for “specific items,” “quantification,” “detailed analysis,” and “visible unmet requirements.”
CMS seed	Generic POC adequacy check.	“Classify whether a CMS Plan of Correction adequately resolves the deficiency.”
CMS GEPA-full	Judge substantial POC adequacy: resident-specific correction, affected population, systemic prevention, staff education, monitoring, timing, accountability, and material omissions tied to the deficiency narrative.	Assess whether the POC provides “clear, actionable steps,” “resident-specific corrections, systemic changes, staff education, and ongoing monitoring,” and avoid relying on “vague or boilerplate language.”

Table 11: Prompt evolution under full feedback. The SEC and CMS optimized prompts share the same context-to-feedback mechanism, but they learn different domain policies. SEC focuses on visible amended disclosure; CMS focuses on whether a remediation plan materially covers the deficiency.

Method	Claim insuff.	Evidence required	Request match	Missing items
GEPA-full	2/2	2/2	2/2	2/2
GEPA-scalar	2/2	2/2	0/2	0/2
GEPA-category	2/2	2/2	0/2	0/2
Req.+action only	2/2	2/2	2/2	2/2

Table 12: Rule presence in SEC optimized prompt candidates. Full feedback more consistently induces named-request matching and missing-item reasoning.

778 prospective-financial-information section. The re-
779 trieved amended evidence gave combined revenues
780 for 2022 and 2023 and some context about who pre-
781 pared the projections. The baseline, GEPA-scalar,
782 and GEPA-category predicted resolved; GEPA-
783 full predicted unresolved.

784 The GEPA-full rationale is qualitatively different.
785 It does not stop at seeing a revised forecast para-
786 graph. It maps the SEC request to named missing
787 requirements: individual-company basis, material
788 assumptions, operating costs, pricing, gross mar-
789 gins, and limitations. This is the regulator-style
790 reasoning we want the system to learn: a response
791 can be responsive in topic but unresolved in obliga-
792 tion coverage.

793 9.4 SEC Case: Stablecoin Definition Without 794 the Requested Contrast

795 In a crypto disclosure case, the SEC asked the com-
796 pany to clarify what it meant by “stable cryptocur-
797 rency” as opposed to an “unstable” cryptocurrency.
798 The company said it changed the term to “stable-
799 coin” and described stablecoins as fiat-pegged as-
800 sets. The amended evidence defined stablecoins

and mentioned examples such as USDC and USDT.
Baseline, MIPROv2, scalar GEPA, and category
GEPA predicted resolved; GEPA-full predicted
unresolved.

The learned policy catches a subtle gap: the evi-
dence defines the stable side but does not directly
provide the requested contrast with unstable cryp-
tocurrency. This is not a lexical trick; it is request
matching. The optimized SEC prompt tells the re-
viewer to identify “specific items or information
requested by the SEC” and verify that amended
evidence covers “all requested aspects.”

9.5 CMS Case: Full Feedback Is Not Just More Skeptical

CMS also shows why full feedback is not merely
an unresolved detector. In one F0689 resident-
safety case, the deficiency was an unsecured treat-
ment cart. The POC gave an immediate teach-
able moment, identified all residents as poten-
tially affected, trained nursing staff on locked
cart storage, reviewed cart storage capacity, and
created weekly/monthly accident-hazard audits
with QAPI review. The generic prompt predicted
not_adequate, arguing that the plan lacked a di-
rect assurance that carts would always be secured.
GEPA-full predicted adequate.

The full feedback for similar adequate cases
teaches calibration: substantial adequacy is enough
when the plan covers immediate correction, af-
fected population, systemic prevention, monitoring,
and timing. This is the opposite of the SEC-style
over-skeptical checklist. It shows that full feed-
back can also reduce false positives by teaching the

reviewer not to demand more than the regulatory object requires.

9.6 CMS Case: Boilerplate Audits Do Not Cover a Multi-Part Deficiency

In another F0689 case, the deficiency narrative combined several issues: fall-risk care planning for a resident, RN eight-hour coverage/PBJ reporting, and refrigerator temperature monitoring. The POC addressed refrigerator gauges, logs, policy education, and audits. GEPA-scalar and GEPA-category predicted adequate. GEPA-full predicted not_adequate, because the POC did not visibly correct the fall-risk care-plan problem or the RN coverage/PBJ problem.

This case is the CMS analogue of the SEC projection example. The response contains plausible compliance language and a real corrective action, but only for one component of the deficiency. Full feedback teaches the model to decompose the deficiency narrative and ask whether each material component has a corresponding corrective action.

9.7 Known Failure Mode

The same mechanism can over-require. In a SEC ownership-percentage case, retrieved evidence contained redemption-scenario calculations and the benchmark label was resolved. GEPA-full predicted unresolved because it wanted a more explicit cover-page presentation. In CMS, the analogous risk is treating a practical POC as incomplete because it lacks idealized audit escalation language. This is why the guarded verifier and CMS adaptation results should be framed as review-assistant policies with calibrated feedback, not as legal or clinical ground truth.

10 Discussion

The main result is not simply that one optimizer beats another. RegTrace-Agent turns a recurring compliance pattern into an offline feedback-learning system: collect the trace, retrieve the revised artifact, check obligations against visible evidence, issue a guarded verdict, and recycle adjudicated failures as feedback. The learned reviewer can be used directly, distilled into a guarded verifier, or converted into lightweight scoring heads for high-throughput triage. The output is a traceable unresolved-gap rationale, not only a binary label.

Language feedback helps because it preserves decision structure. A scalar signal says only that

a verdict failed; full feedback says which named request, company action, amended snippet, and remaining obligation were mismatched. The optimized prompt therefore becomes an interpretable policy: extract material requests, distrust unsupported revision claims, inspect evidence for named items, avoid over-requiring, and distinguish partial disclosure from resolution.

SEC correspondence is the public setting where we can observe request, response, revised filing, and later reviewer behavior, but the trace structure is broader. Audit remediation, regulatory reporting, lending-document stipulations, data-quality exceptions, and internal compliance monitoring also contain request-response-evidence loops. The transferable claim is that such traces can support retrieval, obligation checking, adjudication feedback, optimized reviewing, and hard-case mining; the limitation is that high-recall language-feedback policies can become over-skeptical.

11 Conclusion

We introduced RegTrace-Agent, a context-to-feedback framework for evidence-grounded response review, and instantiated it as RegTrace-SEC using public SEC traces. Across grouped, temporal, and topic-holdout evaluations, full natural language feedback yields the strongest reviewer. Evidence ablations, prompt evolution, and SEC follow-up corroboration suggest that financial NLP benchmarks can move from static labels toward feedback-rich review systems that preserve the reasons behind expert decisions.

Limitations and Data Statement

This study is conditioned on visible retrieved evidence, so missed snippets can affect labels. The SEC benchmark is LLM-adjudicated, with independent audit and external follow-up corroboration but without a full expert legal/accounting audit. A full independent gpt-5.4-mini audit over all 472 examples, using only test-time fields, reaches 0.731 agreement accuracy and 0.715 macro-F1 against the released labels. Same-topic SEC follow-up is a noisy behavioral signal, not ground truth. We frame labels as visible-evidence judgments rather than legal determinations. The CMS-2567 study is weaker and should be read as a cross-regulatory adaptation pilot: its 500 labels are primary LLM adjudications, and the partial boundary is noisier than the SEC visible-evidence label. An anonymized

931 release will be provided at <https://anonymous.>
932 [4open.science/r/regtrace-sec-XXXX/](https://open.science/r/regtrace-sec-XXXX/); it in-
933 cludes benchmark JSONL, splits, accession identi-
934 fiers, retrieval metadata, snippets, labels, feedback
935 fields, prompts, outputs, and analysis scripts.

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1072	A Full Optimized Prompts		
1073	The main paper quotes short excerpts from opti-		
1074	mized prompts. Below are the full GEPA-full op-		
1075	timized candidates exposed by the final grouped-		
1076	split run. The seed prompt was: “Classify whether		
1077	a company resolved an SEC comment using the		
1078	visible evidence.”		
1079	A.1 GEPA-full Optimized Candidate 1		
1080	You are tasked with classifying whether a com-		
1081	pany has resolved a comment from the SEC based		
1082	on the provided evidence. Follow these instruc-		
1083	tions carefully:		
1084	1. Input Format: You will receive three types of		
1085	inputs: <code>sec_comment</code> , <code>company_response</code> , and		
1086	<code>amended_evidence</code> .		
1087	2. Classification Criteria: Determine if the com-		
1088	pany has fully resolved the SEC comment by ana-		
1089	lyzing the company response in conjunction with		
1090	the amended evidence. If the response satisfies		
1091	the SEC’s request and the evidence supports this		
1092	claim, classify the comment as resolved. If the re-		
1093	sponse does not meet the SEC’s request or if there		
1094	are still outstanding issues, classify the comment		
1095	as unresolved.		
1096	3. Evaluation Process: Carefully compare the		
1097	SEC request against the company’s response and		
1098	the amended evidence. Look for specific items or		
1099	information requested by the SEC, quantification		
1100	or detailed analysis that meets SEC requirements,		
1101	references to regulatory guidance or previous de-		
1102	cisions, and clear explanations of how the com-		
1103	pany adjusted disclosures or practices. Take note		
1104	of visible unmet requirements, partial fixes, or		
1105	lack of supporting documentation in the amended		
1106	evidence.		
	4. Output Format: Include a reasoning section,		1107
	a label indicating resolved or unresolved, and a		1108
	concise reason.		1109
	5. Feedback Incorporation: Be vigilant about		1110
	claims made by the company; ensure that		1111
	amended evidence corroborates these claims.		1112
	Identify instances where the response may par-		1113
	tially address SEC concerns but still falls short of		1114
	complete compliance.		1115
	A.2 GEPA-full Optimized Candidate 2		1116
	You are tasked with classifying whether a com-		1117
	pany has resolved a comment issued by the SEC		1118
	based on provided evidence. Follow these specific		1119
	instructions for the task:		1120
	1. Input Format: You will receive <code>sec_comment</code> ,		1121
	<code>company_response</code> , and <code>amended_evidence</code> .		1122
	The SEC comment is the original request for clar-		1123
	ification or additional information. The company		1124
	response indicates what changes were made or		1125
	explanations provided. The amended evidence		1126
	contains snippets from the amended filing or other		1127
	documents that support the response.		1128
	2. Classification Criteria: Determine whether the		1129
	company’s response and amended evidence ade-		1130
	quately address the SEC’s concerns as outlined		1131
	in the original comment. A response is resolved		1132
	if it fully satisfies the requirements of the SEC		1133
	comment, including clear explanations, specific		1134
	details, and quantifications when necessary. A		1135
	response is unresolved if there are still gaps or		1136
	missing elements in the company’s explanation or		1137
	if amended evidence does not address all parts of		1138
	the SEC comment.		1139
	3. Evaluation Process: Carefully analyze the SEC		1140
	comment and identify the specific details or expla-		1141
	nations requested. Review the company response		1142
	to see if the company acknowledged the comment		1143
	and provided a meaningful response. Examine		1144
	amended-evidence snippets to confirm that they		1145
	support the company’s claims and cover all re-		1146
	quested aspects of the SEC comment. Look for		1147
	direct references in the amended evidence that		1148
	address the SEC’s specific requests, such as own-		1149
	ership structures, accounting adjustments, or en-		1150
	forcement risks.		1151
	4. Labeling: If the company has fully addressed		1152
	the SEC’s concerns with adequate evidence, label		1153
	the response resolved. If there are omissions or		1154
	inadequacies in addressing SEC requests, label it		1155
	unresolved.		1156
	5. Justification: Provide clear and concise rea-		1157
	soning that cites specific elements from the SEC		1158
	comment, company response, and amended evi-		1159
	dence. Highlight specific unmet requirements or		1160
	gaps in the evidence that contribute to an unre-		1161
	solved classification.		1162
	6. Feedback Incorporation: Pay attention to feed-		1163
	back from previous examples to identify common		1164
	pitfalls, such as failing to compare the SEC re-		1165
	quest against the evidence adequately or missing		1166
	key elements in the response.		1167
	B CMS Optimized Prompt		1168
	The CMS seed prompt was: “Classify whether a		1169
	CMS Plan of Correction adequately resolves the		1170

deficiency.” The full-feedback CMS run under the 270/90/140 grouped split produced a learned reviewer behavior whose outputs emphasize resident-specific correction, systemic prevention, education, monitoring, timing, and material omissions. We report the final reviewer instruction used for the CMS full-feedback policy below. Unlike the SEC appendix above, this is a paper-facing distilled instruction for the optimized CMS reviewer rather than a raw two-candidate state dump.

Classify the adequacy of a CMS Plan of Correction in resolving identified deficiencies in long-term care facilities. Your evaluation should be based on the specific details of the deficiency and the corresponding Plan of Correction provided. Follow these guidelines:

1. Input Structure: You will receive text inputs including `deficiency_text`, a detailed narrative describing the observed deficiency, including specific incidents and regulatory references; `plan_of_correction_text`, the facility’s proposed corrective actions to address the deficiency; `ftag`, the specific regulatory tag associated with the deficiency; and `scope_severity`, the severity and scope level of the deficiency, indicating its impact on residents.

2. Evaluation Criteria: Assess whether the Plan of Correction provides clear, actionable steps to correct the deficiency. Determine if the Plan includes specific actions addressing resident-specific corrections, systemic changes, staff education, and ongoing monitoring. Identify if the Plan outlines a timeline for implementation and a mechanism for quality assurance or follow-up. Look for any vague or boilerplate language that may undermine the effectiveness of the Plan.

3. Response Types: Classify the Plan of Correction as adequate if it sufficiently addresses the identified deficiencies with clear, actionable solutions and monitoring mechanisms; not adequate if it fails to address the core issues outlined in the deficiency, lacks specificity, or does not include necessary follow-up actions; and partial if it includes some corrective actions but lacks comprehensive strategies or fails to address major components of the deficiency.

4. Justification: Provide a clear and concise rationale for your classification, referencing specific elements from both the deficiency text and the Plan of Correction. Highlight any strengths and weaknesses in the Plan based on the criteria outlined above.

5. Feedback Utilization: Incorporate feedback from previous evaluations to refine your assessment process, ensuring that you identify material omissions and weaknesses tied directly to the deficiency narrative rather than relying solely on generalizable strategies.

C LLM-Assisted TraceSpec Constructor Outputs

The adaptive construction module asks an LLM constructor to propose a domain-specific TRACE-

SPEC from a few examples, then passes the proposal through the executable utility gate. Below are the complete substantive outputs for the three domains. Full JSON artifacts, including token metadata, are released with the code under `outputs/adaptive_trace_framework/llm_schema_proposals/`.

C.1 SEC Constructor Output

Domain: sec.
Task name: visible-evidence response-resolution review.
Decision: go-primary-benchmark.
Role mapping: request=sec_comment; response=company_response; evidence=retrieved_snippets, amended_evidence_best_snippet; label=visible_evidence_resolution_label; feedback=scalar_feedback, category_feedback, full_feedback, feedback_sec_request, feedback_company_action, feedback_evidence_summary, feedback_unmet_requirement; external_corroboration=real_followup_text, has_real_followup; meta-data=review_thread_id, cik, review_file_no, issue_category, response_year.
Label policy: resolved iff the visible amended-filing evidence substantially satisfies the material SEC request; unresolved iff a material requested item, quantification, exhibit, document, legal/accounting analysis, or other obligation is still missing from the visible evidence.
Feedback policy: scalar=binary correctness against the visible-evidence label; category=issue category plus coarse gap/category feedback; full text=natural-language feedback describing the SEC request, company action, visible evidence coverage, and unmet requirement; external text=later same-topic SEC follow-up text, used only as noisy corroboration. Forbidden at test time: feedback fields, full_feedback, amended-evidence adjudication fields, and real_followup_text.
Utility scores: role observability 5.0; evidence availability 4.5; label viability 4.0; feedback richness 5.0; external corroboration 4.0; release viability 4.5; optimization suitability 5.0; mean 4.571.
Risks: visible-evidence labels are not expert legal/accounting review; retrieval-conditioned labels can be affected by missing amended-filing evidence.
Minimal experiment: review a small subset to validate role mapping and feedback richness.
Paper claim: the role mapping and feedback policy align with SEC review because company responses can be evaluated against visible amended-filing evidence and later follow-up.

C.2 CMS Constructor Output

Domain: cms.
Task name: deficiency-to-plan adequacy review.
Decision: go-cross-regulatory-adaptation.
Role mapping: request=deficiency_text;

1427 FDA metadata. Return matched when the can-
 1428 didate closeout is visibly the resolution letter for
 1429 the same FDA warning trace. Strong matching
 1430 signals include the same firm or recipient, same
 1431 CMS/reference number, explicit mention of the
 1432 same warning-letter date, same product or viola-
 1433 tion family, and closeout language saying FDA
 1434 evaluated corrective actions in response to that
 1435 warning. Do not require the closeout letter to re-
 1436 peat every violation from the warning letter; FDA
 1437 closeout letters are often short boilerplate. Return
 1438 mismatched when the candidate closeout appears
 1439 to concern a different firm, different warning date,
 1440 different reference/CMS number, or only shares a
 1441 broad product area.

1442 **D.4 FDA Example: Calibration Repairs** 1443 **Over-Rejection**

1444 One true matched pair involved H2 Beverages, Inc.
 1445 The warning letter was dated June 14, 2022, and
 1446 the candidate closeout letter was dated March 22,
 1447 2023. The strict reviewer rejected the pair because
 1448 the closeout did not repeat enough of the warning-
 1449 letter substance and appeared to use different prod-
 1450 uct wording. The calibrated reviewer predicted
 1451 matched, explaining that the firm identity matched,
 1452 the warning date matched, the product area aligned,
 1453 and the closeout explicitly stated that FDA eval-
 1454 uated corrective actions in response to the same
 1455 warning. This is the FDA analogue of the CMS
 1456 calibration story: a domain reviewer should verify
 1457 trace identity, but it should not over-require a short
 1458 closeout letter to restate every violation.

1459 **E Additional Dataset Details**

1460 The most common issue categories are other disclo-
 1461 sure issues (183), risk factors (91), SPAC disclosure
 1462 (78), revenue recognition (75), crypto (28), non-
 1463 GAAP (16), and cybersecurity (1). The most com-
 1464 mon gap types are resolved visible evidence (153),
 1465 missing specific disclosure (140), missing quantifi-
 1466 cation (59), resolved-but-regulator-requested-more-
 1467 detail (38), missing exhibit or document (29), in-
 1468 complete accounting analysis (25), incomplete le-
 1469 gal or regulatory analysis (25), and other visible
 1470 evidence gap (3).

1471 **F Additional Qualitative Cases**

1472 The qualitative appendix used during analysis in-
 1473 cludes full SEC comments, company responses,
 1474 amended evidence snippets, method predictions,
 1475 GEPA-full rationales, and later SEC follow-up text
 1476 for selected cases. These include stablecoin ter-
 1477 minology, financial forecast assumptions, PIPE
 1478 price-per-share disclosure, Item 404 related-party

disclosure, expense breakout and trend discussion,
 and SPAC transaction-background disclosure. The
 CMS analysis artifacts similarly include deficiency
 narratives, Plans of Correction, covered and miss-
 ing elements, optimizer feedback, and predictions
 from generic, scalar, category, and full-feedback
 reviewers. The two representative CMS cases in
 Section 9 illustrate the two useful directions of
 feedback learning: accepting a materially adequate
 POC despite generic over-skepticism, and rejecting
 a plausible POC that addresses only one component
 of a multi-part deficiency.

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